ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Basic Principles of Computer Programming and Computer Literacy paper 1 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept other correct answers and wording. Flowcharts may be laid out differently, as long as the correct symbols and logic are used.

SECTION A

QUESTION 1

1 1.1 B ✓ 1.2 A ✓ 1.3 D ✓ 1.4 C ✓ 1.5 A ✓ 1.6 B ✓ 1.7 C ✓ 1.8 D ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 False ✓ 2.2 True ✓ 2.3 True ✓ 2.4 False ✓ 2.5 True ✓ (5)

[5]

QUESTION 3

3 3.1 D ✓ 3.2 H ✓ 3.3 F ✓ 3.4 J ✓ 3.5 A ✓ 3.6 C ✓ 3.7 I ✓ 3.8 E ✓ 3.9 B ✓ 3.10 G ✓ (10)

[10]

QUESTION 4

4 4.1 Motion ✓ 4.2 Looks ✓ 4.3 Sound ✓ 4.4 Events ✓ 4.5 Variables ✓ (5)

[5]

TOTAL SECTION A: [30]

SECTION B

QUESTION 5

5.1 5.1.1 Random Access Memory ✓ 5.1.2 Router (modem-router) ✓ 5.1.3 Solid-state drive (SSD) / flash drive / memory card ✓ 5.1.4 Changes the AC mains supply into the low DC voltages that the computer's parts need ✓ (4)

5.2 Hardware is the physical parts of a computer that you can touch ✓ – e.g. keyboard, CPU, hard drive ✓. Software is the programs and instructions that run on the hardware ✓ – e.g. Windows, Python, a web browser ✓. (4)

5.3 A step-by-step set of instructions that solves a problem ✓✓. Any TWO characteristics ✓✓: clear and unambiguous; finite (it ends); correct; has defined inputs and outputs; efficient; steps in the right order. (4)

5.4 A branch is a separate line of development, a copy of the code where changes can be made without affecting the main version ✓✓. Teams use branches so that each person can work on a different feature at the same time ✓, then merge the finished work back into the main branch ✓. (4)

5.5 A variable is a named memory location that stores a value ✓✓. The value of a variable can change while the program runs ✓; a constant's value stays the same ✓. (4)

5.6 $64 + 32 + 0 + 8 + 4 + 0 + 1$ ✓✓✓ $= 109$ ✓✓ (5)

[25]

QUESTION 6

6.1 lives – how many lives the player has left ✓; score – number of correct answers ✓; a – first random number ✓; b – second random number ✓ (4)

6.2 It stops when lives is equal to 0 ✓✓, i.e. when the player has made three wrong answers ✓✓. (4)

6.3 Correct: score increases by 1 ✓ and the sprite says "Correct!" ✓. Wrong: lives decreases by 1 ✓ and the sprite says how many lives are left ✓. (4)

6.4 3 + 4 = 7 correct (score 1); 5 + 5 = 10, answer 9 wrong (lives 2); 2 + 9 = 11, answer 10 wrong (lives 1); 6 + 1 = 7 correct (score 2); 8 + 8 = 16, answer 15 wrong (lives 0) ✓✓. Final message: "Game over! Score: 2" ✓✓ (4)

6.5 Any sensible change ✓✓✓✓, e.g. after "change score by 1" add: if (score mod 5) = 0 then change lives by 1 (and say "Bonus life!"). (4)

[20]

QUESTION 7

7.1 Start terminator ✓ with "Card inserted" / Start ✓ (2)

7.2 Input symbol ✓: "Enter PIN" ✓ (2)

7.3 Decision "PIN correct?" ✓; No path: output "Incorrect PIN" ✓, return card and go to End ✓ (3)

7.4 Yes path: input symbol ✓ "Enter amount" ✓ (2)

7.5 Decision "Amount > balance?" ✓; Yes: output "Insufficient funds" ✓; No: give cash ✓; process "balance = balance − amount" ✓; paths joined to End ✓ (5)

7.6 End terminator ✓ (1)

[15]

QUESTION 8

Any TEN, one mark each ✓✓✓✓✓✓✓✓✓✓: Line 2 – range(1, 5) gives only 4 days, should be range(1, 6); Line 3 – day is a number and must be converted with str(day) before joining; Line 3 – the temperature is not converted to a number (float(input(…))); Line 4 – lists use append(), not add(); Line 5 – total must be the number 0, not the string "0"; Line 6 – temp is not defined, should be temps; Line 9 – missing colon after the if condition; Line 12 – print must be indented inside the elif; Line 14 – missing closing bracket; Line 15 – a string cannot be joined to a number, use str(average) or a comma. (10)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE